Smart Timetable and Campus Resource Coordination System

Introduction

Managing schedules and resources on a busy college campus can be a real challenge. Teachers may be unavailable at the last minute, rooms get double-booked, and students often don't know where they're supposed to be. This proposal introduces a user-friendly, centralized system designed to make campus scheduling smooth, transparent, and efficient for everyone.

Project Overview

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Goal:
To create a single platform—accessible via web and mobile—that keeps everyone on campus informed about class schedules, room assignments, staff substitutions, and event bookings. The system ensures that by 8 AM each day, all schedules and resource allocations are finalized and visible to everyone.

The Problem

- **Teacher Absences:** When teachers are suddenly unavailable, classes are disrupted.
- **Language Hours:** Classes split for language sessions, making room assignments tricky.
- **Event Overlaps:** Events sometimes block classrooms, leading to confusion and last-minute changes.
- **Shared Facilities:** Labs, halls, and sports areas are often double-booked.
- **Lack of Transparency:** Students and staff don't have a clear, real-time view of where they need to be.
- **Inefficient Communication:** Without a centralized system, miscommunication and resource wastage are common.

Our Solution

We propose a cross-platform application that:

- Lets teachers request or accept class substitutions easily.
- Allows language coordinators to assign rooms dynamically for split classes.
- Enables event organizers to check room availability and book spaces without conflicts.
- Gives students and staff a clear, daily view of all room schedules.
- Sends real-time notifications about any changes, with the final schedule locked by 8 AM each day.

Key Features

- **Teacher Substitution Management:** Request and approve class coverages.
- **Language Hour Room Coordination:** Assign rooms for split-language sessions.
- **Event Room Booking:** Book halls, labs, or grounds after checking availability.
- **Real-Time Notifications:** Get instant updates via push and email.
- **Room Availability Dashboard:** See which rooms and facilities are free or booked.
- **Student View:** Access personal and room-wise schedules.
- **Schedule Finalization:** All changes locked and published by 8 AM daily.
- **Audit Trail:** Track changes and resolve conflicts with a clear history.

User Roles & Responsibilities

Role	Capabilities
Teacher	Request/approve substitutions, view schedule
Language Coordinator	Assign/confirm rooms for language classes
Event Organizer	Book rooms/facilities, check availability
Student	View schedules, receive alerts on changes
Non-Teaching Staff	Access facility schedules, assist with logistics/maintenance
Admin (HOD/IT)	Manage users, override bookings, resolve conflicts, view analytics

Supported Room Types

- Classrooms (shared and fixed)
- Language Labs
- Computer/Science Labs

- Library
- Seminar Halls
- Sports Courts (Basketball, Badminton)
- Auditorium
- Playground/Grounds

Technology Stack

Layer	Tool/Framework
Frontend	Flutter (Mobile), React (Web)
Backend	Node.js or Django
Database	Firebase Firestore, PostgreSQL
Auth	Firebase Authentication
Notifications	Firebase Cloud Messaging, Email APIs
Scheduling Logic	Custom time-slot engine, conflict resolution

How the System Works

- **Teachers, Language Coordinators, and Non-Teaching Staff** use the web app to manage schedules and resources.
- **Event Organizers and Students** access the mobile app for bookings and schedule updates.
- All actions are processed by a secure backend, which handles authentication, notifications, schedule management, and resource allocation.
- The database keeps everything in sync and ensures everyone sees the latest information.

User Experience Highlights

- **Teachers:** Request substitutes, view and approve schedule changes.
- **Event Organizers:** Book rooms, see availability, track booking status.
- **Students:** Instantly see their daily schedule and get alerts if anything changes.
- **Language Coordinators:** Assign rooms for language groups on the fly.
- **Admins:** Oversee all bookings, lock the timetable at 8 AM, and resolve conflicts.

Expected Impact

- **No more class or room confusion:** Everyone knows where to be, every day.
- **Efficient resource use:** Facilities are booked and used optimally.
- **Better coordination:** Departments and organizers work together seamlessly.
- **Clear communication:** Students and staff get timely, accurate updates.
- **Less manual work:** Admins and HODs spend less time resolving conflicts.

Conclusion

By bringing all scheduling and resource management into one easy-to-use system, this project will transform how our campus operates. Faculty, students, and staff will benefit from real-time access to schedules, fewer last-minute surprises, and a more organized, efficient academic environment.